

modifications, particularly to the wafer-scanning components that handle precise positioning. The setup would resemble current implementations: the container-sized system sits below the fab floor near the scanner, and mirrors direct the X-ray or XUV light to the wafer.

The largest engineering challenge has been converting what was once a lower-repetition-rate laboratory technology into a continuously operating industrial platform. This requires stable electron bunch generation, precision beam shaping, automated controls, and long-duration operational consistency. A key factor in this transition is increasing the pulse repetition rate of the system. Higher repetition rates directly translate into more exposure events per second, making repetition rate a critical lever for achieving competitive wafer throughput in high-volume manufacturing. The repetition rate works much like a high-speed printer: the more pages it can print per second, the more work it gets done in an hour.

Industrial adoption will depend on reaching competitive high-volume manufacturing throughput targets—roughly 100-250 wafers per hour—and sustaining that performance with maintainable uptime. Reaching this level remains one of the key milestones between prototype success and fab deployment.

The tough road to faster chips

The central challenge in next-generation compact X-ray and EUV sources is generating high-energy electrons inside a small system while maintaining component integrity, beam stability, and industrial reliability. Laser-driven acceleration helps address this by supporting electric field strengths far beyond those achievable in conventional accelerator architectures, enabling rapid acceleration in a much smaller footprint.

Even so, system integration and cost remain major hurdles. While the optical chain may reduce

dependence on some of the large, highly specialised collection optics associated with today's EUV systems, new cost centres emerge in ultra-short-pulse lasers, beamline controls, magnetic systems, stabilisation hardware, and maintenance infrastructure. The long-term economic case will therefore depend on the total cost of ownership rather than on optics-only savings.

Energy efficiency is another promising advantage, particularly through efforts to improve wall-plug-to-X-ray conversion efficiency and recover part of the system's unused energy. While not all losses are practically recoverable, improving this conversion pathway could significantly reduce electricity consumption per wafer over time. Similar to regenerative braking in electric vehicles, some of the wasted energy is captured and reused rather than lost as heat.

The broader industry context also matters. Existing high-NA EUV systems have long deployment lifecycles and enormous installed cost bases, meaning that adoption of any alternative architecture will likely be gradual. Rather than replacing current EUV systems immediately, this approach is strategically positioned to enable the generation of beyond-high-NA tools. As the industry explores even higher numerical apertures, sometimes referred to as hyper-NA, the associated cost and complexity are expected to rise sharply, creating an inflection point where a new light-source architecture becomes more viable.

The strongest case for this next-generation approach lies in its ability to improve performance while avoiding the escalating complexity associated with ever-larger optics and greater dependence on multi-patterning.

Building the future

The platform is designed for scalable deployment, with one compact light source paired to

each scanner, an architecture that aligns well with how current fabs already provision lithography tools. Rather than depending on a massive centralised accelerator feeding multiple scanners, the system can be deployed closer to individual tools, simplifying beam delivery and reducing facility-level integration complexity. Instead of one giant power station serving an entire city block, this works more like placing a dedicated generator next to each building.

A major focus is improving wall-plug efficiency: maximising the conversion of electrical power into usable lithography light at the wafer plane. This includes optimised laser design, reduced optical losses, and partial energy-recovery strategies.

The first near-term applications are outside semiconductor lithography, where commercialisation cycles are shorter, and reliability can be proven faster.

One early target is space radiation testing, where compact accelerator systems can improve access to high-energy particle environments used to validate electronics for orbit. Another is medical therapy, particularly very-high-energy electron-based cancer treatment, where compact accelerator footprints could make advanced treatment systems more affordable than large proton installations.

These adjacent applications serve an important strategic role: they function as real-world validation environments for cost efficiency, power efficiency, uptime, and maintainability. That operational proof can later strengthen the platform's readiness for semiconductor fab integration. In practical terms, these markets serve as real-world proving grounds before the technology is trusted in the semiconductor industry's most demanding production lines. **EFY**

Co-author Nidhi Agarwal is Senior Technology Journalist at EFY.